Dutch Disease or Agglomeration? The Case of Military Expenditures.

Abstract

This paper studies how counties with greater preexisting military exposure respond to changes in federal defense procurement. I build a Bartik-style measure that combines counties' baseline military employment shares with national defense spending and estimate its effects using a staggered difference-in-differences design with continuous treatment. Defense expansions raise employment, income, income per capita, and rents in more exposed counties, but population, military employment, and Republican presidential vote share all decline. Sectoral estimates show little response in manufacturing and gains in agriculture and science and technology services. Overall, defense spending appears to generate local economic benefits through channels beyond a simple expansion of military activity.

1 Introduction

Understanding how defense spending shapes local economies matters for regional development and political economy. Procurement shocks can increase demand, reallocate labor, and crowd out other sectors. Because defense expenditures are often justified partly by their local economic benefits, I investigate whether counties with greater exposure to the military sector respond differently to increases in military procurement.

To evaluate this mechanism, I construct a treatment variable: a Bartik-style measure that interacts defense exposure, measured as the military share of employment, with defense intensity, measured as total real military procurement spending. I then use a difference-in-differences design to estimate its effects on county-level outcomes.

I find that military treatment increases total employment and per capita income. At the same time, contrary to theoretical predictions, population and military-sector employment decline, which calls for further investigation. Interestingly, Republican presidential vote share also falls in more highly treated counties.

The policy implications remain inconclusive because the effects move in different directions across outcomes. Still, the strength of these results suggests that important mechanisms remain unexplored and merit further study.

This paper is most closely related to Allcott and Keniston (2018), who study whether U.S. natural resource booms generate Dutch disease or instead stimulate broader local activity through agglomeration and input–output linkages. More broadly, my project builds on the Dutch disease framework of Corden and Neary (1982) and on the long-run

evidence in Michaels (2011), which shows that oil abundance reshaped local specialization and development in the U.S. South. Relative to this literature, I study a different source of local shocks: federal military procurement rather than natural resource booms.

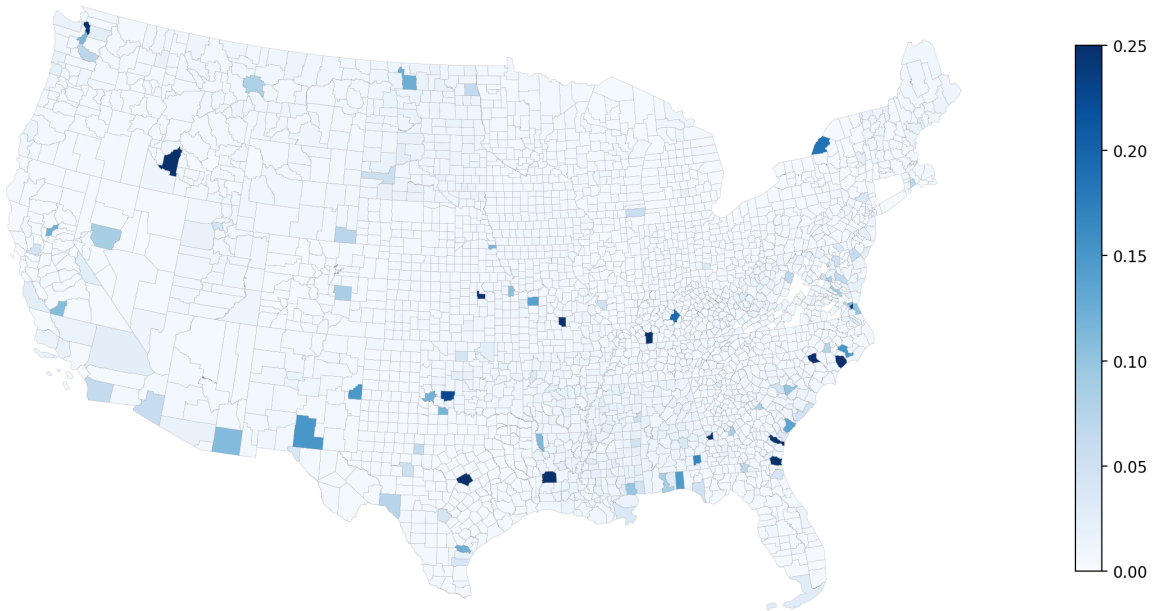
The project also connects to work on military spending and regional economic activity, such as Hooker and Knetter (1997), and to broader political-economy analyses of defense spending, such as Griffin et al. (1982) and Nincic and Cusack (1979). My contribution is to bring these strands together in a county-level design: I estimate how military spending shocks interact with pre-existing defense exposure and trace their effects on employment, income, population, and voting.

The remainder of the draft proceeds as follows: Section 2 describes the setting I exploit and the data I use. Section 3 describes the results, and Section 4 concludes.

2 Setting and Data

I follow Allcott and Keniston (2018) closely in both the theoretical framing and the empirical setup, replacing resource booms with military expansions. The core idea is that, in the short run, increases in military spending should have stronger local effects in communities that are more specialized in defense-related activity. I examine whether an expansion in military procurement increases population, wages, local employment, and housing prices, and whether it reduces employment in tradable sectors such as manufacturing.

Figure 1: Exposure. Share of Military Employment



This figure shows the share of employment in the military sector in each county in 2001. For visualization, exposure is truncated at 0.25. In the full sample, the mean is 0.012, the standard deviation is 0.03, the minimum is 0.0007, and the maximum is 0.887.

Most of the county-level data come from the Census and the Regional Economic Information System. I obtain county population, total personal income, and per capita income from table CAINC1, and sectoral employment from table CAEMP25N. I take national military expenditure data from the OMB Historical Tables (Table 3.2, "Outlays by Function"), county-level election outcomes from Data and Lab (2018), and housing data from the American Community Survey for 2000, 2010, 2015, and 2021.

Most of the datasets I use begin in 2001 and end in 2022, in part because some earlier Census tables were discontinued or redefined. In principle, analogous series exist for earlier years and could be used to extend the panel. In practice, however, matching industry codes consistently across the major coding changes around 2000 proved difficult within the scope of this project, so I retain the shorter sample. This limited time span is an important constraint of the analysis.

I construct a county-level exposure measure as

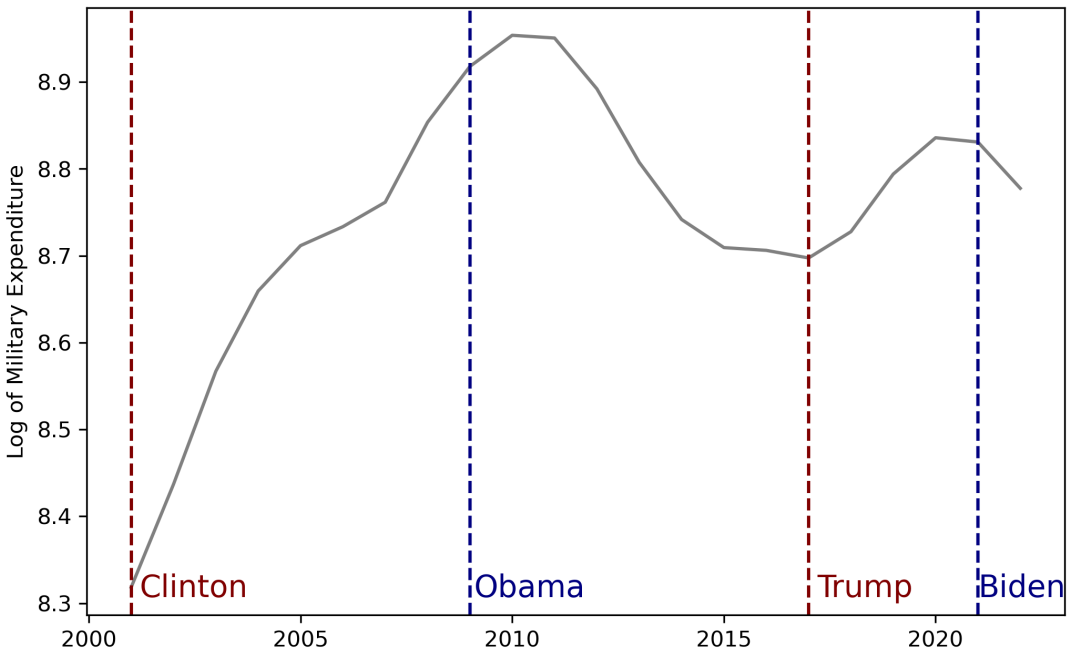
$$Exposure_i = \frac{Military_Employment_{i,2001}}{Total_Employment_{i,2001}},$$

where military employment includes both defense production and military services. I then define a Bartik-style treatment variable as

$$Bartik_{it} = Exposure_i \cdot Military_Expenditures_t,$$

which serves as the main regressor throughout the analysis. In some specifications, I trim this variable at 0.05 to facilitate imputation. As a robustness check, I also replace the level of military expenditures with its logarithm when constructing the treatment.

Figure 2: Magnitude. National Military Expenditures



This figure shows log military expenditures over time. The mean is 8.74 and the SD is 0.15.

Figure 1 maps the exposure measure across counties. Most counties have military employment shares close to zero, while the relatively small number of highly exposed counties stand out clearly. The average exposure is about 1.2%, with substantial dispersion across the sample. Figure 2 plots military expenditures over time. The mean of the logged series is 8.74, with a standard deviation of 0.15. The series suggests that real military procurement tends to rise during Republican administrations and to grow more slowly, or decline, during Democratic administrations.

Table 1: Descriptives. Outcomes

	Military Share			
	Low (<1%)	High	Total	p-value
Log Population	10.323	10.126	10.269	0.000
	(1.520)	(1.340)	(1.475)	
Log Employment	9.679	9.310	9.577	0.000
	(1.537)	(1.424)	(1.516)	
Log Employment in Military	4.479	5.049	4.636	0.000
	(1.484)	(1.826)	(1.606)	
Log Income	13.907	13.635	13.832	0.000
	(1.602)	(1.428)	(1.561)	
Log Income per Capita	10.491	10.417	10.471	0.000
	(0.316)	(0.324)	(0.320)	
Log Median Housing Rent	6.365	6.302	6.347	0.000
	(0.418)	(0.468)	(0.433)	
Share of Votes for Republican Candidate	0.606	0.588	0.601	0.000
	(0.152)	(0.157)	(0.154)	
Number of Observations	49.588	19.008	68.596	

This table reports descriptive statistics for the outcome variables. Military share is measured as the share of county employment in the military sector in 2001.

Table 1 summarizes the main outcome variables. A simple difference-in-means test indicates that counties with high and low levels of military employment differ significantly across all outcomes considered here. Counties with higher military shares tend to be smaller, have lower per capita income, and vote less Republican. Appendix B provides figures on the growth rates of the main outcome variables over time.

3 Results

A common empirical approach in the literature is to use Bartik-style measures and difference-in-differences estimators to identify treatment effects. I use the estimator from De Chaisemartin and d'Haultfoeuille (2024) to account for heterogeneous treatment effects, which is one of the most appropriate estimators when treatment is staggered and

continuous. In particular, for each treated unit, we can estimate the effect

$$DiD_{il} = Y_{i,t+l}(D_1) - Y_{it}(D_0) - \frac{1}{N_j}[Y_{j,t+l}(D_0) - Y_{jt}(D_1)]$$

where Y is the outcome, unit i switches from D_0 to D_1 in $t+1$, all counties in j remain at that treatment level, and DiD_{il} is the unit-specific l-period treatment effect. To obtain a non-zero number of stayers, I trim the treatment variable for values close to zero. The individual effects are then aggregated using weights that depend on the number of treated units and periods, as well as on treatment intensity. For details, see De Chaisemartin and d'Haultfoeuille (2024).

Table 2: Effects of Defense Expansions

	LPop	LEmp	LMEmp	LInc	LIncpc	LRent	RepShare
Treatment	−0.320	2.441	−0.748	3.029	3.284	0.923	−3.031
	(0.080)	(0.158)	(0.237)	(0.212)	(0.230)	(0.248)	(0.294)
Observations	65264	65264	65242	65242	65242	9228	12241

Columns report DiD results. The outcomes are Log Population, Employment, Military Employment, Income, Income per Capita, and Rent, as well as the Share of Votes for the Republican Candidate. Treatment is a Bar-Tick indicator equal to Exposure (Share of Military in Total Employment) multiplied by Real Expenditure on National Defense trimmed at a value of 0.05. Standard errors are clustered at the county level.

Table 2 reports the main results. Employment increases with treatment intensity. A county with a baseline military share of 10% experiences a 0.24 log-point increase in population for each additional million dollars spent on defense by the federal government, which corresponds roughly to 24%. The effects on employment, income, and income per capita are also positive, with coefficients of approximately 0.30 and 0.33 log points for the latter two outcomes. However, both population and military employment *decrease* in counties with higher initial military shares, which runs counter to the theoretical prediction. This inconsistency may warrant further investigation. As in Allcott and Keniston (2018), I also find that rents increase in places with higher treatment. Once again, somewhat unexpectedly, the share of votes for Republican candidates—who typically support military expansions—decreases by 3 percentage points.

Table 3 shows the effects on other sectors. Manufacturing is not affected by treatment, which could either imply that there is no connection or mask opposing effects between upstream manufacturing and non-linked tradable manufacturing, as in Allcott and Keniston (2018). Notably, agricultural employment increases, as does employment in science and technology services. All of the above results are fairly robust to alternative specifications, such as OLS or two-way fixed effects, as well as to alternative data manipulations, such as trimming at different levels or on different variables. Appendix A reports the results

of these robustness checks.

Table 3: Effects on Alternative Employment

	Log Manu Emp	Log Agr Emp	Log Educ Emp	Log Science Emp
Treatment	0.376	1.320	0.038	1.721
	(0.581)	(0.214)	(1.123)	(0.544)
Observations	57518	64029	40824	43785

Columns report DiD results. The outcomes are Log Employment in Manufacturing, Agriculture, Education, and Science or Technology. Treatment is a Bar-Tick indicator equal to Exposure (Share of Military in Total Employment) multiplied by Real Expenditure on National Defense trimmed at a value of 0.05. Standard errors are clustered at the county level.

The decrease in military employment alongside an increase in science employment in response to higher military expenditures appears counterintuitive. One possible explanation is that, in the 21st century, defense expenditures may be more focused on research, development, and weapons production than on hiring additional soldiers. This could also help explain why lower-rank workers previously employed by the military appear to reduce their support for Republicans. However, this interpretation remains speculative and should be explored in future work.

4 Conclusion

My project studies how counties with greater military exposure respond to increases in federal defense procurement. Using a Bartik-style variable and a staggered difference-in-differences approach, I find that defense expansions raise employment, income, income per capita, and rents in more exposed counties. Yet population, military employment, and Republican vote share all decline, suggesting that defense spending affects local economies through channels beyond a simple expansion of military activity.

Sectoral results reinforce this interpretation: manufacturing shows little response, while agriculture and science and technology services gain employment. It might be important to examine the composition of defense spending more closely and trace the mechanisms linking it to labor reallocation, migration, and politics in future work.

References

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A Additional Tables

B Additional Figures

Table 4: OLS on Exposure

	LPop	LEmp	LMEmp	LInc	LIncpc	LRent	RepShare
Military Share	3.397	3.853	18.078	3.336	−0.061	0.948	−0.235
	(0.283)	(0.239)	(0.751)	(0.293)	(0.030)	(0.126)	(0.039)
Constant	10.230	9.533	4.421	13.793	10.471	6.335	0.604
	(0.007)	(0.006)	(0.010)	(0.007)	(0.001)	(0.004)	(0.001)
Num. Observations	68374	68374	68351	68374	68374	12317	15307
Mean Dep. Variable	10.271	9.579	4.638	13.833	10.470	6.347	0.601
Adj. R-squared	0.005	0.006	0.114	0.004	0.000	0.004	0.002

The outcomes are Log Population, Employment, Military Employment, Income, Income per Capita, and Rent, as well as the Share of Votes for the Republican Candidate. Exposure is the Share of Military in Employment.

Table 5: OLS Results

	LPop	LEmp	LMEmp	LInc	LIncpc	LRent	RepShare
Treatment	0.389	0.441	2.066	0.385	−0.004	0.121	−0.027
	(0.032)	(0.027)	(0.086)	(0.034)	(0.003)	(0.015)	(0.004)
Constant	10.230	9.533	4.421	13.793	10.471	6.334	0.604
	(0.007)	(0.006)	(0.010)	(0.007)	(0.001)	(0.004)	(0.001)
Num. Observations	68374	68374	68351	68374	68374	12317	15307
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All columns report OLS results. The outcomes are Log Population, Employment, Military Employment, Income, Income per Capita, and Rent, as well as the Share of Votes for the Republican Candidate. Treatment is a Bar-Tick indicator equal to Exposure (Share of Military in Total Employment) multiplied by Real Expenditure on National Defense. Standard errors are clustered at the county level.

Table 6: TWFE Results

	LPop	LEmp	LMEmp	LInc	LIncpc	LRent	RepShare
Treatment	0.078	0.668	0.408	0.364	0.286	0.353	−0.573
	(0.305)	(0.152)	(0.294)	(0.301)	(0.094)	(0.124)	(0.109)
Constant	10.263	9.509	4.595	13.795	10.440	6.310	0.661
	(0.032)	(0.016)	(0.031)	(0.032)	(0.010)	(0.013)	(0.011)
Num. Observations	68374	68374	68351	68374	68374	12313	15307
Mean Dep. Variable	10.271	9.579	4.638	13.833	10.470	6.347	0.601
Adj. R-squared	0.998	0.997	0.995	0.997	0.944	0.946	0.824

All columns report TWFE results. The outcomes are Log Population, Employment, Military Employment, Income, Income per Capita, and Rent, as well as the Share of Votes for the Republican Candidate. Treatment is a Bar-Tick indicator equal to Exposure (Share of Military in Total Employment) multiplied by Real Expenditure on National Defense. Standard errors are clustered at the county level.

Table 7: Robustness. Trimming at Bar-Tick Level of 0.1

	LPop	LEmp	LMEmp	LInc	LIncpc	LRent	RepShare
Treatment	−0.169	0.724	−0.192	1.194	1.362	1.044	−0.185
	(0.056)	(0.095)	(0.217)	(0.124)	(0.139)	(0.190)	(0.133)
Observations	65242	65242	65242	65242	65242	9228	12241

Columns report DiD results. The outcomes are Log Population, Employment, Military Employment, Income, Income per Capita, and Rent, as well as the Share of Votes for the Republican Candidate. Treatment is a Bar-Tick indicator equal to Exposure (Share of Military in Total Employment) multiplied by Real Expenditure on National Defense trimmed to 0 at 0.1. Standard errors are clustered at the county level.

Table 8: Robustness. Trimming by Exposure

	LPop	LEmp	LMEmp	LInc	LIncpc	LRent	RepShare
Treatment	0.052	3.981	−0.638	4.866	4.813	1.645	−5.185
	(0.114)	(0.246)	(0.337)	(0.323)	(0.338)	(0.357)	(0.424)
Observations	65242	65242	65242	65242	65242	9228	12241

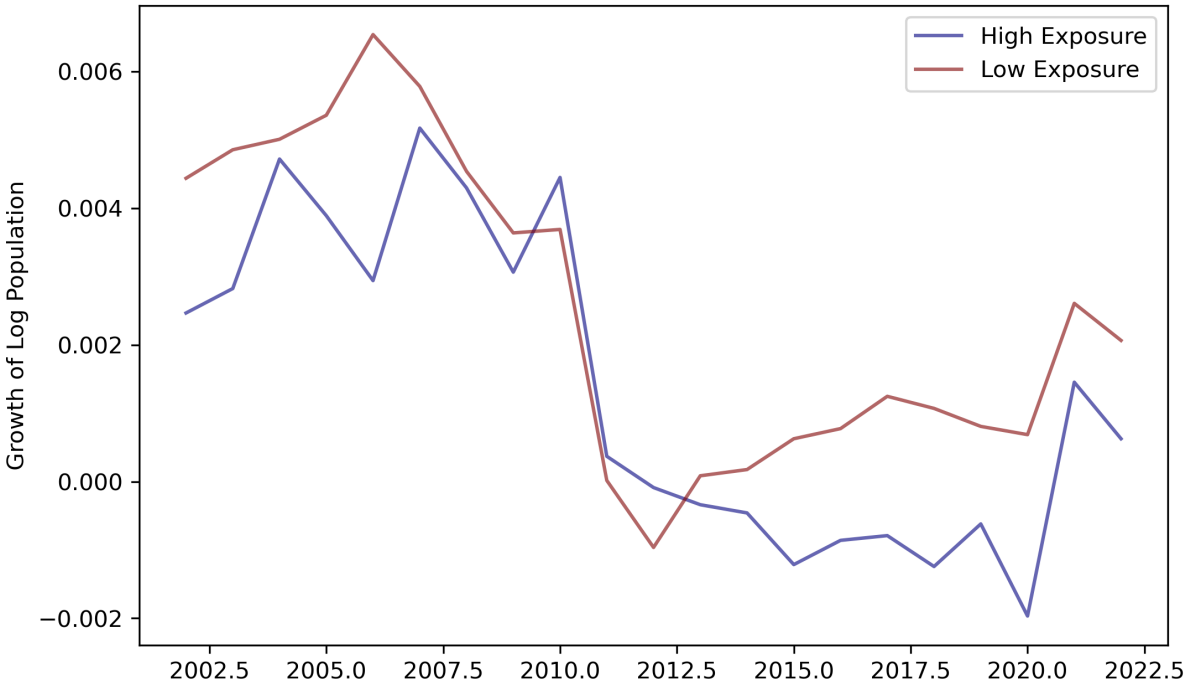
Columns report DiD results. The outcomes are Log Population, Employment, Military Employment, Income, Income per Capita, and Rent, as well as the Share of Votes for the Republican Candidate. Treatment is a Bar-Tick indicator equal to Exposure (Share of Military in Total Employment) multiplied by Real Expenditure on National Defense trimmed to 0 for all exposure levels below 0.005. Standard errors are clustered at the county level.

Table 9: Robustness. Trimming by Exposure

	LPop	LEmp	LMEmp	LInc	LIncpc	LRent	RepShare
Treatment	−0.095	4.036	−0.638	5.160	5.255	1.645	−5.196
	(0.114)	(0.248)	(0.337)	(0.332)	(0.351)	(0.357)	(0.424)
Observations	65264	65264	65242	65264	65264	9228	12245

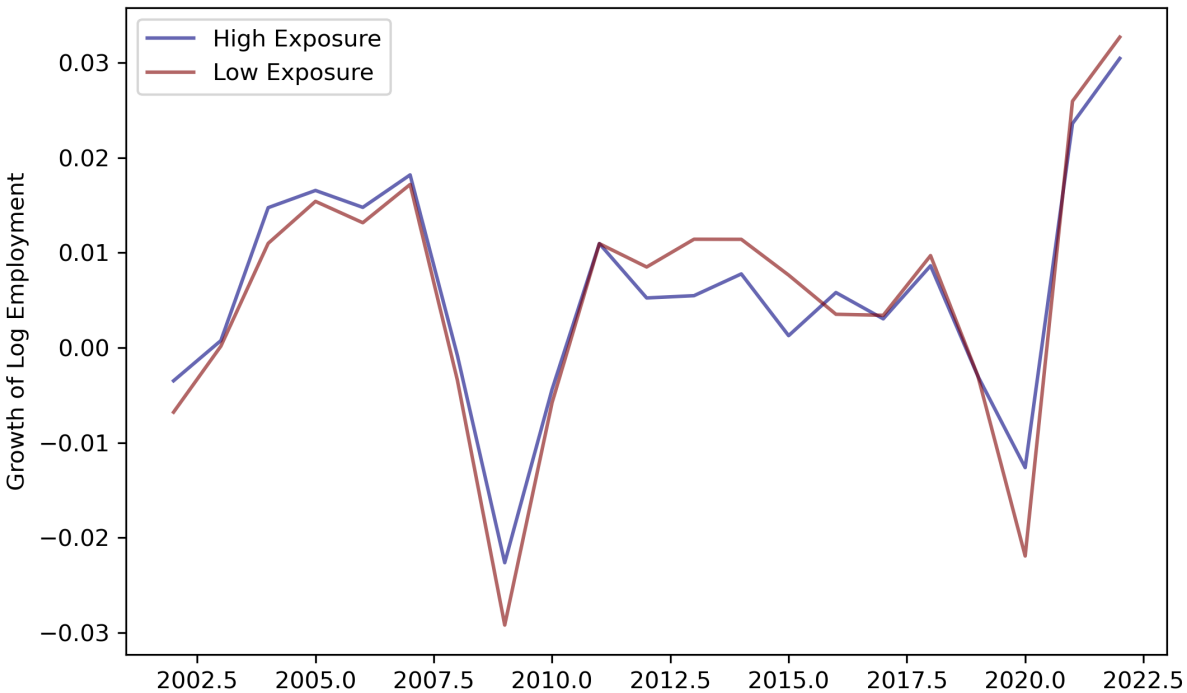
Columns report DiD results. The outcomes are Log Population, Employment, Military Employment, Income, Income per Capita, and Rent, as well as the Share of Votes for the Republican Candidate. Treatment is a Bar-Tick indicator equal to Exposure (Share of Military in Total Employment) multiplied by Log Real Expenditure on National Defense trimmed as in Table 8. Standard errors are clustered at the county level.

Figure 3: Population Growth Rate



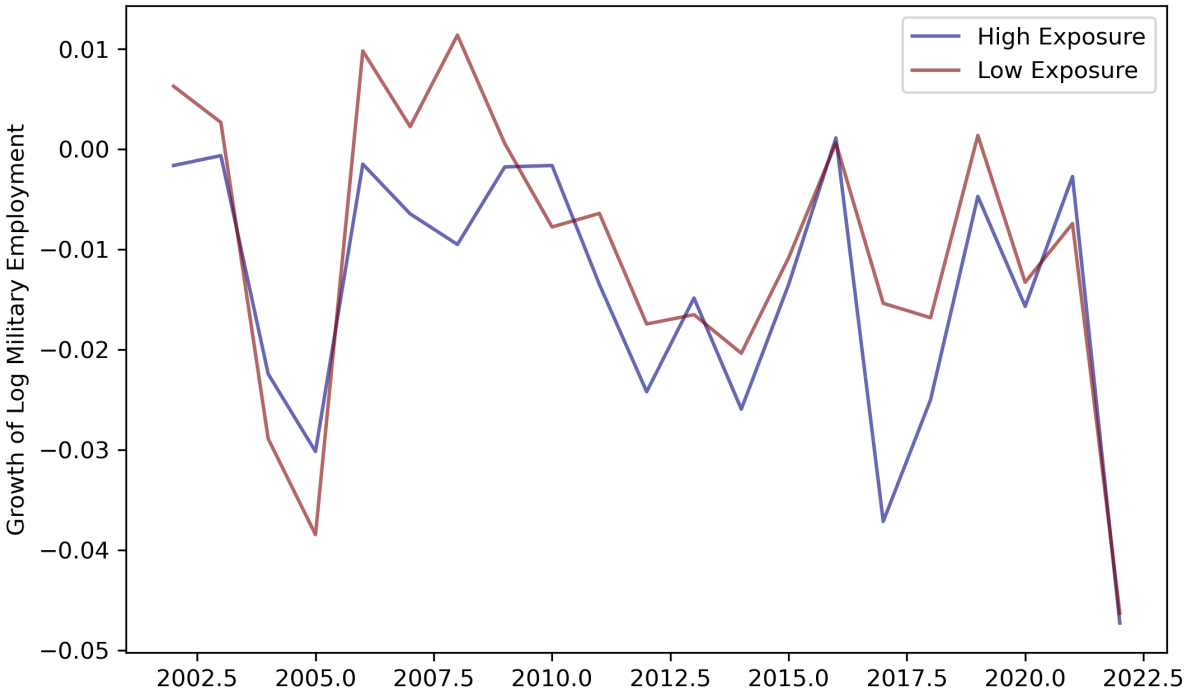
Note: This figure shows the mean growth rate of population over time.

Figure 4: Employment Growth Rate



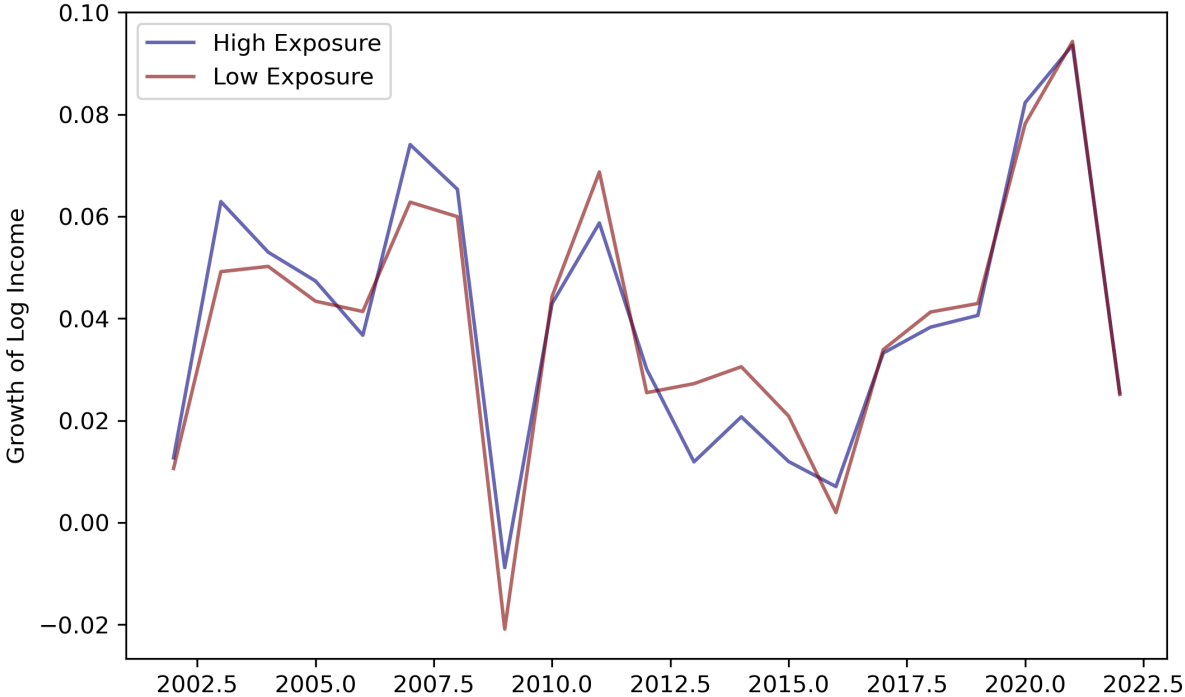
Note: This figure shows the mean growth rate of employment over time.

Figure 5: Military Employment Growth Rate



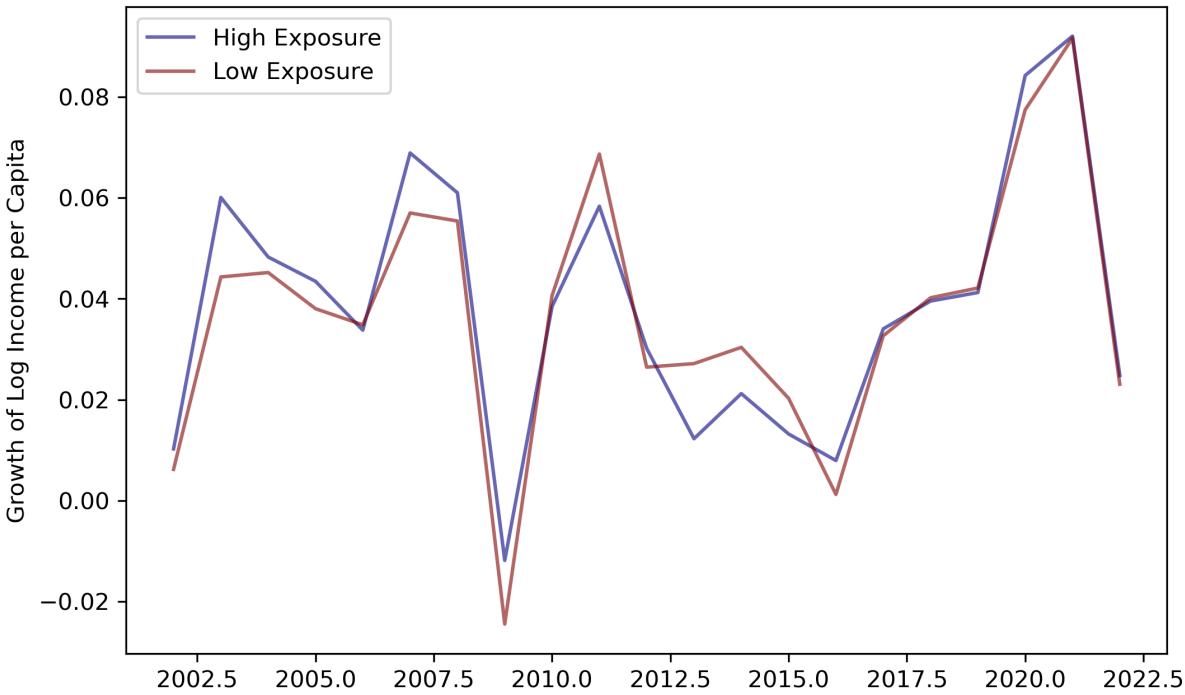
Note: This figure shows the mean growth rate of military employment over time.

Figure 6: Income Growth Rate



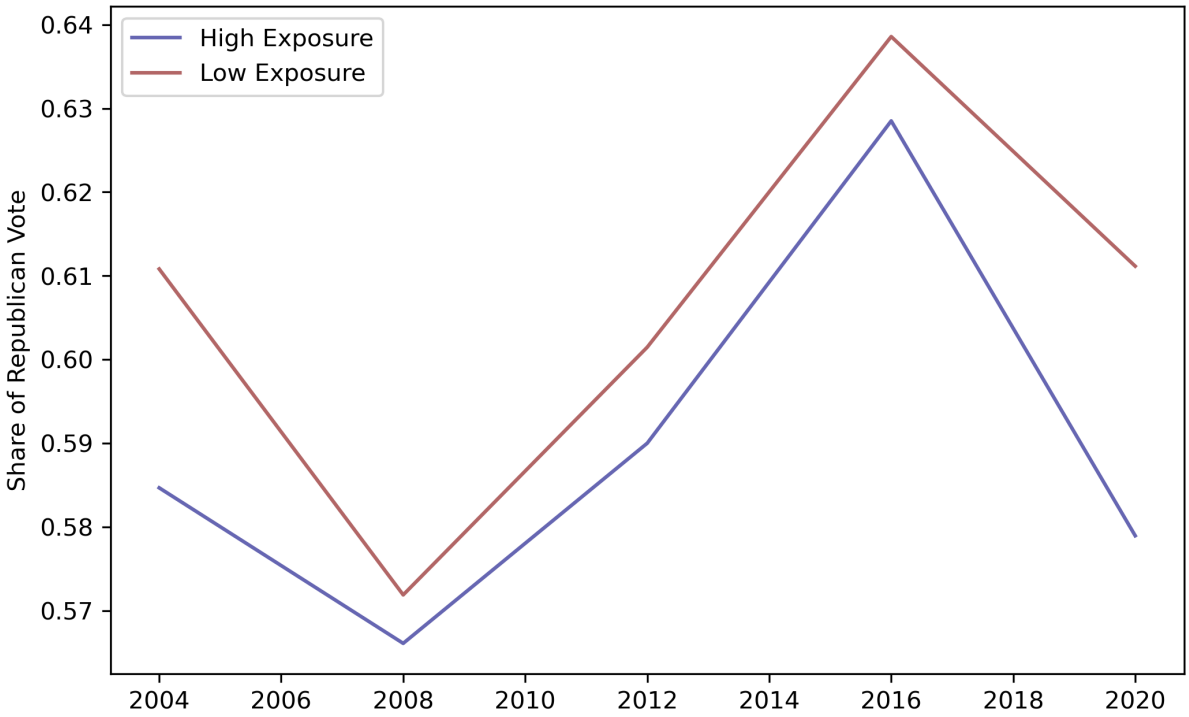
Note: This figure shows the mean growth rate of personal income over time.

Figure 7: Income per Capita Growth Rate



Note: This figure shows the mean growth rate of income per capita over time.

Figure 8: Share of Votes for Republican Candidate



Note: This figure shows the mean share of votes for the republican candidate over time.